

Biostat STR Generation 3 500 L - DeltaV, dual 3-blade configuration

sartorius-str-gen3-500

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	125 L	Source-supported
Maximum working volume	500 L	Source-supported
Nominal vessel / bag size	500 L	Source-supported
Total vessel volume loaded into the form	700 L	Source-supported
Initial working volume loaded into the form	125 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	1.23	Source-supported
Internal / chamber diameter	0.815 m	Source-supported
Liquid height at maximum working volume	1 m	Source-supported
Impeller / mixing type	Hydrofoil	App assumption
Impeller count	2	Source-supported
Impeller diameter	0.31 m	Source-supported
Power number	0.4	App assumption
Initial agitation	80 rpm	App assumption
Maximum agitation	110 rpm	Source-supported
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.2 vvm	Source-supported
Baffled vessel	No	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 125 L

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 125 L.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

[Local original: Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7](#)

[Manufacturer publication](#)

Maximum working volume - 500 L

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

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Nominal vessel / bag size - 500 L

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

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Total vessel volume loaded into the form - 700 L

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 700 L.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

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Initial working volume loaded into the form - 125 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(125 L, 500 L) = 125 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 1.23

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.23.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

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Internal / chamber diameter - 0.815 m

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.815 m.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

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Liquid height at maximum working volume - 1 m

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1 m.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

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[Manufacturer publication](#)

Impeller / mixing type - Hydrofoil

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].impellerType = "hydrofoil"`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption.

Impeller count - 2

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 6; Table 4 footnote 1

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.

Limitations: Selected 2 x 3-blade impellers, not mixed 3/6-blade configuration.

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[Manufacturer publication](#)

Impeller diameter - 0.31 m

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7; Table 6, 500 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.31 m.

Limitations: Flexsafe STR geometry. Direct dimensions take precedence over rounded ratios; bag configuration can affect immersion volumes.

[Local original: Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 7](#)

[Manufacturer publication](#)

Power number - 0.4

App assumption | Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 5; Process engineering characterization, printed p. 18

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0.4.

Limitations: De Wilde 2014 PDF p. 5 reports torque-based Ne about 1.3 for the TWO three-blade segment impeller configuration in turbulent flow ($Re > 10000$), not an unambiguous per-impeller coefficient. Retained app coefficient is unverified; do not silently apply configuration-level Ne to each impeller or transfer it to mixed impellers.

Local original: [Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 5](#)

Initial agitation - 80 rpm

App assumption | Large mammalian-vessel scaling rule

Rule: $baseRpm = maxVolume \geq 1000 \text{ L} ? 55 : 80$

Selection / calculation: 500 L is not at least 1000 L -> 80 rpm

Limitations: The simulator lowers its generic starting agitation at larger scale. This is an internal scale heuristic, not a manufacturer limit. This is the initial agitation setting loaded into the form.

Maximum agitation - 110 rpm

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 4; Table 4 maximum stirrer speed

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 110 rpm.

Limitations: Valid for dual 3-blade configuration; mixed-impeller limits may be lower depending on fill.

Local original: [Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 4](#)

[Manufacturer publication](#)

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: $PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.2 vvm

Source-supported | Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 4; Table 4 maximum total gassing rate

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.2 vvm.

Limitations: Total gassing limit. Micro-sparger-only limit is 0.1 vvm.

Local original: [Sartorius Biostat STR Generation 3 Emerson datasheet - PDF page 4](#)

[Manufacturer publication](#)

Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: $PROFILE_DEFAULTS["mammalian"].baffled = false$

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].maxPressure = 0.5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].operatingPressure = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

Standard large STR Generation 3 configurations are primarily cell-culture oriented; the simulator uses a generic mammalian stirred-tank profile.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Reviewed De Wilde et al., Superior Scalability of Single-Use Bioreactors, BPI September 2014, PDF pp. 1-6 (printed 14-19).

Manufacturer-authored characterization article, not a current hardware datasheet. Archived source:

sartorius-dewilde-scalability-2014-27b44451dc52.pdf

2014 Table 1 corroborates current selected absolute vessel/impeller diameters and liquid height. Current Generation 3 datasheet remains controlling; historical 25% minimum fills, sparger details and process-study speeds do not override current limits. Article text p. 4 says vessel H/D 2:1 whereas Table 1 lists 1.8; neither is substituted for liquid H/D.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Supplementary archived source: Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 2

Historical comparison only; not blanket validation of current model

Local original document identities

Sartorius Biostat STR Generation 3 Emerson datasheet

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Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014

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